Project Title

Full-Text Search Engine on a Suffix-Array Index

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in
A.Y. 2026-2027

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ABSTRACT

The Full-Text Search Engine on a Suffix-Array Index is an advanced information retrieval system designed to provide fast, accurate, and efficient searching over a large collection of text documents. Traditional search methods often require scanning entire documents, which becomes slow and inefficient as the size of the dataset increases. To overcome this limitation, the proposed system constructs a Suffix Array, a compact and memory-efficient indexing structure that enables rapid searching of text. The search engine supports substring, prefix, and exact phrase queries, allowing users to retrieve relevant information with high accuracy. The project implements an $O(n \log^2 n)$ suffix array construction algorithm and utilizes Kasai's Longest Common Prefix (LCP) algorithm to efficiently compute the common prefixes between adjacent suffixes. Furthermore, LCP-accelerated binary search is employed to reduce unnecessary character comparisons during query processing, significantly improving search performance and minimizing response time.

The system also incorporates a ranking mechanism that orders search results based on their relevance, ensuring that the most useful documents appear first. To evaluate the effectiveness of the proposed approach, the suffix-array-based search engine is compared with a traditional inverted index by analyzing factors such as query latency, index size, memory consumption, and search efficiency. The comparison highlights the strengths and trade-offs of both indexing techniques while demonstrating the advantages of suffix arrays for substring and phrase searching. This project showcases the practical application of advanced string-processing algorithms in building scalable and high-performance search systems. It can be applied to various real-world domains, including digital libraries, search engines, document management systems, code repositories, e-book platforms, and bioinformatics, where fast and accurate text retrieval is essential.

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